Jan Janssen

Max Planck Institute for Sustainable Materials
Max-Planck-Str. 1, 40237 Düsseldorf, Germany

Email: janssen@mpi-susmat.de
Website: jan-janssen.com

PROFESSIONAL APPOINTMENTS

2023 – present **Group leader for Materials Informatics** - Computational Materials Design
Research topic: foundation models for sustainable materials
Max Planck Institute for Sustainable Materials, Düsseldorf, Germany

2026 – present **Guest Lecturer** - Interdisciplinary Centre for Advanced Materials Simulation (ICAMS)
Ruhr University Bochum, Bochum, Germany

2024 **Invited Fellow** - Institute for Mathematical and Statistical Innovation (IMSI)
Research topic: data-driven materials informatics
University of Chicago, Chicago, IL, USA

2022 – 2023 **Director's Postdoctoral Fellowship** - Theoretical Division (T-1)
Research topic: long-timescale simulations of materials with *ab initio* accuracy
Los Alamos National Laboratory, Los Alamos, NM, USA

2023 **Invited Postdoctoral Fellow** - Institute for Pure and Applied Mathematics (IPAM)
Research topic: new mathematics for the exascale: applications to materials science
University of California, Los Angeles, CA, USA

2021 – 2022 **Postdoctoral Research Associate** - Theoretical Division (T-1)
Research topic: exascale atomistic capability for accuracy, length, and time
Los Alamos National Laboratory, Los Alamos, NM, USA

2017 **Invited Fellow** - Institute for Pure and Applied Mathematics (IPAM)
Research topic: complex high-dimensional energy landscapes
University of California, Los Angeles, CA, USA

2015 – 2021 **PhD Candidate** - Computational Materials Design
Research topic: computational phase studies and *ab initio* thermodynamics
Max-Planck-Institut für Eisenforschung, Düsseldorf, Germany

EDUCATION

2015 – 2021 **PhD in Theoretical Physics** - Paderborn University, Germany
- Thesis: pyiron - an integrated development environment for *ab initio* thermodynamics
- Advisor: Prof. Jörg Neugebauer
- Grade: summa cum laude

2009 – 2015 **Advanced Degree in Physics** - Technical University of Kaiserslautern, Germany
- Thesis: carbon in α-iron-grainboundaries - an atomistic study of elastic properties
- Advisor: Prof. Herbert Urbassek
- Grade: thesis 1.0 (excellent) - total 1.6 (good)

FUNDED PROPOSALS

2026 **Principal Investigator** as part of the National Research Data Infrastructure for Materials Science and Engineering (NFDI-Matwerk)
- Funding: EUR 190k

2024 **Principal Investigator** as part of the International Max Planck Research School for Sustainable Metallurgy - from Fundamentals to Engineering Materials
- Funding: EUR 150k

Co-Principal Investigator as part of the collaborative research center (CRC) 1394 "Structural and chemical atomic complexity - from defect phase diagrams to material properties"
- Funding: EUR 160k

2023 **Principal Investigator** of "workflows for machine learned interatomic potentials" as part of an internal call for funding for Postdoctoral Fellows at Los Alamos National Laboratory
- Computing Hardware Funding: USD 13k

Co-Principal Investigator of "uncertainty propagation for multi-fidelity machine learned interatomic potentials" as part of the summer school for applied machine learning at Los Alamos National Laboratory
- Funding: USD 25k

2022 **Co-Principal Investigator** of "development and deployment of a fully autonomous in *silico* processing and materials discovery platform" in collaboration with the department of Mechanical Engineering at Texas A&M University
- Funding: USD 176k

Co-Principal Investigator of "transferability of interatomic machine learning potentials" as part of the summer school for applied machine learning at Los Alamos National Laboratory
- Funding: USD 25k

Principal Investigator of "helping users build workflows with ontological typing" project funded by a NumFOCUS small development grant to hire an independent contractor
- Funding: USD 6k

AWARDS & HONORS

2024 Winner of the "2nd Annual Large Language Model Hackathon for Applications in Materials Science and Chemistry" coupling a large language model to atomistic simulation

Invited fellow at the Institute for Statistical and Mathematical Innovation (IMSI)

2023 Invited postdoctoral fellow at the Institute for Pure and Applied Mathematics (IPAM)

2022 Los Alamos National Laboratory (LANL) director's postdoctoral fellowship

2021 PhD with highest distinction - summa cum laude

2019 Runner-up for the Heinz Billing Award - a national biennial award for outstanding contributions to computational science by younger scientist without tenure

2017 Invited fellow at the Institute for Pure and Applied Mathematics (IPAM)

2015 Scholarship of the Technical University of Kaiserslautern

PUBLICATIONS

Google Scholar, 10/26: Citations: 746

2026 **J. Janssen**, O. Waseda, B. Bayerlein, S. Stier, S. Brinckmann, A.A. Guzmán, S. Menon, M.A. Popp, J.R. Pedersen, I.E. Castelli, J. Waitelonis, W. Wenzel, J. Neugebauer, J. Schaarschmidt and T. Hickel. Toward Full Interoperability in Materials Science: Integrating Workflows With Knowledge Graphs. *Adv. Eng. Mater.* doi:10.1002/adem.71258. Citations: 0

M. Bruns, L. Abdul, S. Barth, M. Glauer, M. Günther, F. Neuhaus, J.-O. Perschewski, T. Bjarsch, H. Abel, J. Bosch, N. Meyer, P. Hofmann, M. Drechsler, A. Dyck, D.K. Nerella, M. Tegeler, O. Shchyglo, I. Steinbach, T. Klecker, H. Birkholz, E. B. Farahani, B. Fedelich, C. Haftaoglu, T. Lakshmipathy, E.G. Trelles, C. Schweizer, R.D. Kamachali, L. Koschmieder, J. Eiken, M. Apel, S.-M. Wi, J. Zhao, A. Fezer, N. Christ, P. Schneider, E. Urinov, V. Königer, V. Knoblauch, J. Harter, T. Seifert, T. Janpheng, F. Fuchs, F. Teichert, J. Schuster, A. Bardehle, A. Limper, U. Giese, C. Lucke, M. Beiner, C. Gögelein, L. Leuchtenberger-Engel, C. Hopmann, F. Wentzien, H. Wittek, B. Klie, A. Röhrs, L. Overmeyer, A. Atila, M. Sadowski, L. Talirz, **J. Janssen**, J.R. Pedersen, P. Beck, T. Vegge, I.E. Castelli, W. Wenzel, T. Hickel and J. Schaarschmidt Supporting AI Readiness Through Digital Workflows in Materials Science. *Adv. Eng. Mater.* doi:10.1002/adem.71262. Citations: 1

J. Janssen, and J. Neugebauer. pysqa - Python Simple Queuing System Adapter. *Journal of Open Source Software*. doi:10.21105/joss.10961. Citations: 0
- Code: pyiron/pysqa

V.F. Grizzi, P.J. Lohr, N. Fedik, S. Anaya, J. Averitt, J. Gomes, Y. Wai Li, K. Barros, N. Lubbers, R.A. Messerly, **J. Janssen**, M. Kulichenko, J. Smith, S. Tretiak and B.T. Nebgen. ALF: Open-Source Active Learning Framework for Atomistic Modeling. *J. Chem. Theory Comput.*. doi:10.1021/acs.jctc.6c01122. Citations: 1
- Code: lanl/ALF

S. Brinckmann, S. Menon, R. Röske, J. Steinhülb, M. Selzer, **J. Janssen**, T. Hickel, R. Schwaiger and J. Neugebauer. A Lightweight Procedural Layer for Hybrid Experimental-Computational Workflows in Materials Science. *Adv. Eng. Mater.* doi:10.1002/adem.202503176. Citations: 2

Sk Md A.A. Alvi, B. Vela, V. Attari, **J. Janssen**, D. Perez, D. Allaire, and R. Arróyave. Deep Gaussian process-based cost-aware batch Bayesian optimization for complex materials design campaigns. *npj Computational Materials*. doi:10.1038/s41524-026-01981-7. Citations: 7

M. Stricker, L. Banko, N. Sarazin, N. Siemer, **J. Janssen**, L. Zhang, J. Neugebauer and A. Ludwig. Computationally accelerated experimental materials characterization - drawing inspiration from high-throughput simulation workflows. *npj Computational Materials*. doi:10.1038/s41524-025-01919-5. Citations: 6

I. Baghishov, **J. Janssen**, G. Henkelman and D. Perez. Application-specific Machine-Learned Interatomic Potentials: Exploring the Trade-off Between DFT Convergence, MLIP Expressivity, and Computational Cost. *Digital Discovery*. doi:10.1039/D5DD00294J. Citations: 8
- Code: IlgarBaghishov/DFTprecisionMLIP

2025

Sk Md A.A. Alvi, M. Mulukutla, N. Flores, D. Khatamsaz, **J. Janssen**, D. Perez, D. Allaire, V. Attari and R. Arróyave. Accurate and uncertainty-aware multi-task prediction of HEA properties using prior-guided deep Gaussian processes. *npj Computational Materials*. doi:10.1038/s41524-025-01811-2. Citations: 7
- Code: sheikhahnaf/DGP_with_EncoderDecoderPrior

J. Janssen, J. George, J. Geiger, M. Bercx, X. Wang, C. Ertural, J. Schaarschmidt, A.M. Ganose, G. Pizzi, T. Hickel and J. Neugebauer. A Python workflow definition for computational materials design. *Digital Discovery*. doi:10.1039/D5DD00231A. Citations: 18
- Code: pythonworkflow/python-workflow-definition

Y. Zimmermann, A. Bazgir, A. Al-Feghali, M. Ansari, L.C. Brinson, Y. Chiang, D. Circi, M.-H. Chiu, N. Daelman, M.L. Evans, A.S. Gangan, J. George, H. Harb, G. Khalighinejad, S.T. Khan, S. Klawohn, M. Lederbauer, S. Mahjoubi, B. Mohr, S.M. Moosavi, A. Naik, A.B. Ozhan, D. Plessers, A. Roy, F. Schöppach, P. Schwaller, C. Terboven, K. Ueltzen, S. Zhu, **J. Janssen**, C. Li, I. Foster, B. Blaiszik. 32 examples of LLM applications in materials science and chemistry: towards automation, assistants, agents, and accelerated scientific discovery. *Machine Learning: Science and Technology*. doi:10.1088/2632-2153/ae011a. Citations: 50
- Code: jan-janssen/LangSim

J. Janssen, M.G. Taylor, P. Yang, J. Neugebauer and D. Perez. Executorlib – Up-scaling Python workflows for hierarchical heterogenous high-performance computing. *Journal of Open Source Software*. doi:10.21105/joss.07782. Citations: 7
- Code: pyiron/executorlib

Sk Md A.A. Alvi, **J. Janssen**, D. Khatamsaz, D. Perez, D. Allaire and R. Arróyave. Hierarchical Gaussian process-based Bayesian optimization for materials discovery in high entropy alloy spaces. *Acta Materialia*. doi:10.1016/j.actamat.2025.120908. Citations: 33
- Code: sheikhahnaf/DGP-BO

2024

J. Janssen, E. Makarov, T. Hickel, A.V. Shapeev and J. Neugebauer. Automated optimization and uncertainty quantification of convergence parameters in plane wave density functional theory. *npj Computational Materials*. doi:10.1038/s41524-024-01388-2. Citations: 14
- Code: pyiron/pyiron-dft-uncertainty

T.J. Summers, M.G. Taylor, L.J. Augustine, **J. Janssen**, D. Perez, E.R. Batista, and P. Yang. On the Importance of Configuration Search to the Predictivity of Lanthanide Selectivity. *JACS Au*. doi:10.1021/jacsau.4c00770. Citations: 23

S. Menon, Y. Lysogorskiy, A. L. M. Knoll, N. Leimeroth, M. Poul, M. Qamar, **J. Janssen**, M. Mrovec, J. Rohrer, K. Albe, J. Behler, R. Drautz, and J. Neugebauer. From electrons to phase diagrams with classical and machine learning potentials automated workflows for materials science with pyiron. *npj Computational Materials*. doi:10.1038/s41524-024-01441-0. Citations: 39
- Code: pyiron/potential_publication

M. Tynes, M.G. Taylor, **J. Janssen**, D. Burrill, D. Perez, P. Yang, and N. Lubbers. Linear graphlet models for accurate and interpretable cheminformatics. *Digital Discovery*. doi:10.1039/D4DD00089G. Citations: 10
- Code: lanl/minervachem

2023 M.G. Taylor, D.J. Burrill, **J. Janssen**, E. Batista, D. Perez, and P. Yang. Architector: high-throughput cross-periodic table 3D complex builder. *Nature Communications*. doi:10.1038/s41467-023-38169-2. Citations: 72
- Code: lanl/Architector

A. Rohskopf, C. Sievers, N. Lubbers, M. A. Cusentino, J. Goff, **J. Janssen**, M. McCarthy, D. Montes de Oca Zapiain, S. Nikolov, K. Sargsyan, E. Sikorski, L. Williams, D. Sema, A. P. Thompson, and M. A. Wood. FitSNAP: Atomistic machine learning in LAMMPS. *Journal of Open Source Software*. doi:10.21105/joss.05118. Citations: 71
- Code: FitSNAP/FitSNAP

2021 L.F. Zhu, **J. Janssen**, S. Ishibashi, F. Körmann, B. Grabowski and J. Neugebauer. A fully automated approach to calculate the melting temperature of elemental crystals. *Computational Materials Science*. doi:10.1016/j.commatsci.2020.110065. Citations: 63
- Code: pyiron/pyiron_meltingpoint

2020 T.D. Swinburne, **J. Janssen**, M. Todorova, G. Simpson, P. Plechac, M. Luskin and J. Neugebauer. Anharmonic free energy of lattice vibrations in fcc crystals from a mean field bond. *Physical Review B*. doi:10.1103/PhysRevB.102.100101. Citations: 9
- Code: tomswinburne/BLaSA

2019 **J. Janssen**, S. Surendralal, Y. Lysogorskiy, M. Todorova, T. Hickel, R. Drautz and J. Neugebauer. pyiron: an integrated development environment for computational materials science. *Computational Materials Science*. doi:10.1016/j.commatsci.2018.07.043. Citations: 216
- Code: pyiron/pyiron

Y. Lysogorskiy, T. Hammerschmidt, **J. Janssen**, J. Neugebauer and R. Drautz. Transferability of interatomic potentials for molybdenum and silicon. *Modelling and Simulation in Materials Science and Engineering*. doi:10.1088/1361-651X/aafd13. Citations: 27

2016 **J. Janssen**, N. Gunkelmann and H. M. Urbassek. Influence of C concentration on elastic moduli of $\alpha' - Fe_{1-x}C_x$ alloys. *Philosophical Magazine*. doi:10.1080/14786435.2016.1170224. Citations: 13

PUBLICATIONS IN REVIEW

2025 Z. Wang, H. Huang, H. Zhao, C. Xu, S. Zhu, **J. Janssen** and V. Viswanathan. DREAMS: Density Functional Theory Based Research Engine for Agentic Materials Simulation. doi:10.48550/arXiv.2507.14267. Citations: 49
- Code: BattModels/material_agent

BOOK CHAPTERS

2026 T. Hickel, **J. Janssen**, S. Menon, O. Waseda, L. Huber and J. Neugebauer. pyiron - Developing and Managing Materials Science Workflows. Published in *Workflow systems for large-scale scientific data analysis*, Berlin Universities Publishing.

OPEN-SOURCE SOFTWARE

2015 – present **pyiron** - an integrated development environment for computational materials science
- Role: Lead developer for a team of eight core developers
- Code: pyiron

2018 – present **Conda-forge** - community-led software distribution for the conda package manager
- Role: Maintainer for materials science software
- Contribution: over 1000 packages with a total of over 500 million downloads
- Code: conda-forge/staged-recipes

2026 **pysqa** - Python Simple Queuing System Adapter
- Contribution: Lead the development
- Code: pyiron/pysqa

2025 **executorlib** - Up-scale python functions for high performance computing (HPC)
- Contribution: Lead the development
- Code: pyiron/executorlib

2024 **LangSim** - Large language model for atomistic simulation
- Contribution: Lead the team of international researchers
- Code: jan-janssen/LangSim

2022 **Architector** - high-throughput cross-periodic table 3D complex builder
- Contribution: Parallelization of the chemical complex building using mpi4py
- Code: lanl/Architector

2021 **FitSNAP** - Software for generating SNAP machine-learning interatomic potentials
- Contribution: Implemented a python library interface for the Exascale computing project
- Code: fitsnap/fitsnap

WORKSHOPS

2026 **Organizer** for the Minisymposium on "Agentic Workflows for Trustworthy Discovery in Materials Science and Chemistry" as part of the *Platform for Advanced Scientific Computing conference*, Bern, Switzerland

2025 **Co-organizer** for the workshop on "Machine Learning" as part of the International Max Planck Research School for Sustainable Metallurgy. *ICAMS, Ruhr University*, Bochum, Germany

2021 **Co-organizer** for the workshop on "workflows for atomistic simulation".
Max-Planck-Institut für Eisenforschung, Düsseldorf, Germany (online)
- Code: pyiron-workshop/potentials-workshop-2021

2020 **Co-organizer** for the workshop on "software tools from atomistics to phase diagrams".
Pennsylvania State University, Pennsylvania, USA (online)
- Code: pyiron-workshop/phasediagram-workshop-2020

TEACHING

2026 **Shared lecture series with Prof. G. Sutmann** on "Programming and Numerical Methods" for bachelor students in materials science and simulation. *Ruhr University Bochum*, Bochum, Germany.

Two lectures on "Extension of Density Functional Theory: From Molecules to Solids" as part of the lecture on "Molecular Photonics and Excited-State Processes (MPESP)" for master students in chemistry. *Heinrich Heine University*, Düsseldorf, Germany.

Single lecture on "pyiron - Complex Workflows Made Easy" for NFDI-MatWerk Summer School. *Ruhr University Bochum*, Bochum, Germany.

2025 **Shared lecture series with Prof. G. Sutmann** on "Programming Concepts in Materials Science" for master students in materials science and simulation. *Ruhr University Bochum*, Bochum, Germany.

Single lecture on "From Atoms to Material Properties: An introduction to ab-initio thermodynamics" as part of the lecture on "Materials Modelling" for master students in materials science. *Birmingham University*, Birmingham, United Kingdom.

Single lecture on "Extension of Density Functional Theory: From Molecules to Solids" as part of the lecture on "Molecular Photonics and Excited-State Processes (MPESP)" for master students in chemistry. *Heinrich Heine University*, Düsseldorf, Germany.

2024 **Single lecture** on "Density Functional Theory" as part of the International Max Planck Research School for Sustainable Metallurgy lecture series on Thermodynamics. *Max Planck Institute for Sustainable Materials*, Düsseldorf, Germany

2023 **Single lecture** on "Introduction to Materials Informatics" as part of the Graduate Autumn School of the CRC1349. *Rheinisch-Westfälische Technische Hochschule (RWTH)*, Aachen, Germany

SUPERVISION

2025 – present **Prabhath Chilakalapudi** (PhD student): Machine-learned interatomic potentials combining theoretical predictions and experimental references
- Role: Supervisor - acquiring the funding and leading the supervision of the PhD student

2024 – present **Haitham Wael Ali Awad Gaafer** (PhD student): Machine-learned interatomic potentials for chemo-mechanics balancing computational efficiency and accuracy
- Role: Supervisor - acquiring the funding and leading the supervision of the PhD student

2026 – present **Varun Nair** (Master Student): Genetic Algorithms for high-throughput screening of defects
- Role: Supervisor - for a student research project

2024 – 2026 **Ziqi Wang** (PhD student): Autonomous discovery of novel battery materials with large language model agents and machine-learned interatomic potentials.
- Role: Co-Supervisor - in collaboration with Prof. Viswanathan at University of Michigan

2023 – 2026 **Sk Md Ahnaf Akif Alvi** (PhD student): Autonomous in silico processing and materials discovery for additive manufacturing using Bayesian optimization.
- Role: Co-Supervisor - in collaboration with Prof. Arroyave at Texas A&M University

2023 – 2026 **Ilgar Baghishov** (PhD student): Multi-fidelity machine-learned interatomic potentials
• Role: Co-Supervisor - in collaboration with Prof. Henkelman at the University of Texas at Austin

PRESENTATIONS

INVITED TALKS

2026 **J. Janssen**. Large Language Model Agents for Materials Research Workflows. *Physical Sciences Data Infrastructure & Royce Materials Data Summit Workshop*, Manchester, United Kingdom.

J. Janssen. Large Language Model Agents Towards Autonomous Materials Discovery. *CECAM Workshop: Open Databases Integration for Materials Design*, Grenoble, France.

J. Janssen. Autonomous Materials Discovery Using the pyiron Workflow Framework *Machine Learning in Chemical and Materials Sciences Workshop*, Santa Fe, NM, USA.

J. Janssen. FAIR data management with the pyiron workflow framework, *CECAM Workshop: FAIR Data management of theoretical spectroscopy and Green's function methods*, Lausanne, Switzerland.

J. Janssen. Towards Autonomous Materials Design with the pyiron Workflow Framework, *Seminar series organized by Materials Valley e.V.*, Hanau, Germany (online).

J. Janssen. Uncomfortable Truth about Materials Informatics, *Group Seminar at University of Stuttgart*, invited by Prof. B. Grabowski, Stuttgart, Germany.

J. Janssen. Scalable sustainable materials design with the pyiron workflow framework, *Seminar at National Energy Research Scientific Computing (NERSC)*, Berkeley, CA, USA.

J. Janssen. Autonomous, Data-Driven Workflows for Materials Acceleration Platforms with pyiron, *DPG Spring Conference*, Dresden, Germany.

J. Janssen. Up-scaling Python functions for HPC with executorlib, *Seminar series organized by Workflows Community*, Chicago, IL, USA (online).

2025 **J. Janssen**. Autonomous Materials Science Research, *Artificial Intelligence Symposium on Theory, Application, and Research (AI STAR) Workshop at European Space Agency*, Darmstadt, Germany.

J. Janssen. Can a machine predict novel sustainable materials? Towards using LLMs for Materials Design, *Seminar at Birmingham University*, invited by Prof. Nils Warnken, Birmingham, United Kingdom.

J. Janssen. Autonomous Materials Science Research, *Seminar at German Research Centre for Artificial Intelligence*, Saarbrücken, Germany.

J. Janssen. Up-scale Python Functions for High-Performance Computing with Executorlib, *Seminar series organized by Department of Energy Python Exchange*, New York, NY, USA (online).

J. Janssen. Predict the chemistry of sustainable materials with machine learning, *Seminar at University of Utah*, invited by Prof. A. Clark, Salt Lake City, UT, USA.

J. Janssen. Accelerating Materials Discovery with Machine Learning, *Nanomaterials: Computation, Theory, Machine Learning and Experiment Workshop*, Telluride, CO, USA.

J. Janssen. Large Language Model Agents Towards Autonomous Materials Discovery, *Seminar at the Center for Nonlinear Studies (CNLS) at Los Alamos National Laboratory*, Los Alamos, NM, USA.

J. Janssen. Machine Learned Interatomic Potentials for Extreme Environments, *Workshop on Models and Data for Plasma-Material Interactions in Fusion*, Vienna, Austria.

J. Janssen. pyiron – Workflows for data-driven Materials Science, *NHR4CES Community Workshop*, Darmstadt, Germany (online).

J. Janssen. Predicting Sustainable Materials with Machine Learning, *Machine learning with computational materials science data for modelling nanocrystal catalysts workshop*, Freiburg, Germany.

2024 **J. Janssen**. How to use machine learning in the discovery and design of materials for the future?, *Seminar at Michigan Institute for Computational Discovery & Engineering*, invited by Prof. V. Viswanathan, Michigan, MI, USA.

J. Janssen. Foundation Models for Materials Discovery, *Summer School at Center for Scientific Foundation Models*, Michigan, MI, USA (online).

J. Janssen. pyiron: Simulation Workflows for Data-Driven Materials Design, *Seminar at the Center for Nonlinear Studies (CNLS) at Los Alamos National Laboratory*, Los Alamos, NM, USA.

J. Janssen. pyiron: workflows for the development and assessment of interatomic potentials, *Workshop on Machine Learning Force Fields at the institute for mathematical and statistical innovation (IMSI)*, Chicago, IL, USA.

2023 **J. Janssen** and J. Neugebauer. Rapid prototyping and up-scaling atomistic workflows with pyiron, *Ab initio Description of Iron and Steel (ADIS) Workshop: Digitalization and Workflows*, Kreuth, Germany.

J. Janssen. Packaging Scientific Software with Conda-Forge, *Workshop at the Institute for Pure and Applied Math (IPAM)*, Los Angeles, CA, USA.

J. Janssen and D. Perez. Up-scaling atomistic simulation workflows with pyiron, *Workshop at the Institute for Pure and Applied Math (IPAM)*, Los Angeles, CA, USA.

J. Janssen, M. G. Taylor, P. Yang and D. Perez. Tutorial on high-throughput screening for chemical science with pyiron, *2nd International Workshop on Theory Frontiers in Actinide Science : Chemistry & Materials*, Santa Fe, NM, USA.

2022 **J. Janssen**, T. Hickel and J. Neugebauer. Automated atomistic calculation of thermodynamic and thermophysical data, *DPG Fall conference*, Regensburg, Germany.

J. Janssen. pyiron - an integrated development environment (IDE) for scientific workflows at scale, *Seminar series organized by the Department of Energy Python Exchange*, New York, NY, USA (online).

J. Janssen. Up-scaling simulation protocols with pyiron, *Workshop organized by the Platform Material Digital*, Berlin, Germany (online).

J. Janssen. pyiron - an integrated development environment (IDE) for materials science, *Special Interest Group Data Infrastructure (SIGDIUS) Seminar at Stuttgart University*, Stuttgart, Germany (online).

2021 **J. Janssen** and J. Neugebauer. pyiron - an integrated development environment for materials science, *CECAM Workshop: Simulation Workflows in Materials Modelling (SWiMM)*, Lausanne, Switzerland (online).

2020 **J. Janssen**, T. Hickel and J. Neugebauer. Uncertainty quantification for *ab initio* thermodynamics, *Seminar at Ruhr University Bochum*, invited by Prof. R. Drautz, Bochum, Germany (online).

J. Janssen, T. Hickel and J. Neugebauer. Automated *ab initio* determination of materials properties at finite temperatures with pyiron, *NIST Workshop: Atomistic simulations for industrial needs*, Gaitgersburg, MD, USA (online).

2019 **J. Janssen**, T. Hickel and J. Neugebauer. Automated *ab initio* determination of materials properties at finite temperatures with pyiron, *Seminar at the Center for Nonlinear Studies (CNLS) at Los Alamos National Laboratory*, Los Alamos, NM, USA.

J. Janssen, T. Hickel and J. Neugebauer. pyiron - an integrated development environment for computational materials science, *Seminar at University of Vienna*, invited by Prof. G. Kresse Vienna, Austria.

TALKS AT INTERNATIONAL CONFERENCES

2026 **J. Janssen** and J. Neugebauer. Large Language Model Agents For Atomistic Simulation Workflows, *Materials Science and Engineering Conference*, Darmstadt, Germany.

J. Janssen. pyiron – A Workflow Framework for Trustworthy Agentic Workflows in Materials Science, *Platform for Advanced Scientific Computing*, Bern, Switzerland.

T. Hickel, **J. Janssen** and J. Neugebauer. Data-Driven and LLM-Based Design of Hydrogen Solubilities in Metallic Alloys, *TMS Spring Meeting*, San Diego, CA, USA.

J. Janssen and J. Neugebauer. Large Language Model Agents for Atomistic Simulation Workflows, *TMS Spring Meeting*, San Diego, CA, USA.

J. Janssen, P. Chilakalapudi, H. Gaafer and J. Neugebauer. Uncertainty Propagation for Ab Initio Thermodynamic Phase Diagrams, *TMS Spring Meeting*, San Diego, CA, USA.

H. L. Mai, **J. Janssen**, T. Hickel and J. Neugebauer. Accelerating grain boundary segregation studies in ferritic steels with machine learned interatomic potentials, *DPG Spring Meeting*, Dresden, Germany.

J. Janssen. Leveraging AI in Simulation Workflows for Sustainable Materials Design with pyiron, *deRSE Conference*, Stuttgart, Germany.

2025 **J. Janssen**. Up-scaling Python functions for HPC with executorlib, *Supercomputing Conference*, St. Louis, MO, USA.

J. Janssen and J. Neugebauer. Automated Workflows for Fitting Machine Learned Interatomic Potentials on the Exascale with pyiron, *Psi-k conference*, Lausanne, Switzerland.

J. Janssen and J. Neugebauer. Large Language Model Agents for Atomistic Simulation Workflows, *MSE Research Data Forum*, Siegburg, Germany.

J. Janssen and J. Neugebauer. Inverse Materials Design with Large Language Models, *DPG Spring Meeting*, Regensburg, Germany.

2024 **J. Janssen**, J. Neugebauer and D. Perez. Potential in a day - up-scale the fitting of machine learning potentials for the Exascale, *IPAM Workshop*, Los Angeles, CA, USA.

J. Janssen and J. Neugebauer. pyiron - Simulation Workflows for the Design of Sustainable Materials, *MRS Fall Meeting*, Boston, MA, USA.

M. Stricker, L. Banko, N. Sarazin, **J. Janssen**, N. Siemer, J. Neugebauerand A. Ludwig. Computationally Accelerated Experimental Materials Characterization Using pyiron, *Multiscale Materials Modelling (MMM) conference*, Prague, CZ.

J. Janssen and J. Neugebauer. Automated Workflows to Construct and Validate Machine-Learned Interatomic Potentials, *Multiscale Materials Modelling (MMM) conference*, Prague, CZ.

I. Baghishov, **J. Janssen**, G. Henkelman and D. Perez. Investigating the uncertainty in multi-fidelity machine learning interatomic potentials, *TMS Spring Meeting*, Orlando, Fl, USA.

J. Janssen and D. Perez. Transferable Machine Learning Potentials for Extreme Environments, *TMS Spring Meeting*, Orlando, FL, USA.

2023 **J. Janssen** and D. Perez. Parameter studies for interatomic potentials using LAMMPS and pyiron, *LAMMPS Virtual Workshop and Symposium 2023*, Philadelphia, PA, USA (online).

J. Janssen and D. Perez. Enabling long timescale molecular dynamics simulation with ab initio precision, *MRS Spring Meeting*, San Francisco, CA, USA.

J. Janssen and D. Perez. pyiron an integrated development environment for the development and assessment of interatomic models, *11th Annual Mach Conference*, Baltimore, MD, USA.

J. Janssen and D. Perez. Enabling long timescale molecular dynamics simulation with ab initio precision, *TMS Spring Meeting*, San Diego, CA, USA.

2022 **J. Janssen**, T. Hickel and J. Neugebauer. Predicting melting temperatures from bulk properties with pyiron, *Multiscale Materials Modelling (MMM) conference*, Baltimore, MD, USA.

J. Janssen, M. G. Taylor, P. Yang and D. Perez. Screening of ligand-metal-complexes for separation science with pyiron, *ACS Fall Meeting*, Chicago, IL, USA.

J. Janssen. pyiron – an integrated development environment (IDE) for scientific workflows, *Scientific Computing with Python (SciPy)*, Austin, TX, USA.

J. Janssen. Predicting melting temperatures from bulk properties with pyiron, *Artificial Intelligence for Materials Science (AIMS)*, Gaithersburg, MD, USA (online).

2019 **J. Janssen**, T. Hickel and J. Neugebauer. Automated uncertainty quantification for *ab initio* thermodynamics, *MRS Fall Meeting*, Boston, MA, USA.

J. Janssen, T. Hickel and J. Neugebauer. Automated sensitivity analysis for high-throughput *ab initio* calculations, *IPAM Workshop*, Los Angeles, CA, USA.

J. Janssen, T. Hickel and J. Neugebauer. Automated error analysis and control for *ab initio* calculations, *DPG Spring Meeting*, Regensburg, Germany.

J. Janssen, T. Hickel and J. Neugebauer. Automated sensitivity analysis for high-throughput *ab initio* calculations, *TMS Spring Meeting*, San Antonio, TX, USA.

2018 **J. Janssen**, T. Hickel and J. Neugebauer. Generation of *ab initio* datasets with predefined precision using uncertainty quantification, *DPG Spring Meeting*, Berlin, Germany.

2017 **J. Janssen**, T. Hickel and J. Neugebauer. Towards an uncertainty quantification for *ab initio* thermodynamics, *MRS Fall Meeting*, Boston, MA, USA.

J. Janssen, T. Hickel and J. Neugebauer. Sensitivity analyis for large sets of density functional theory calculations, *DPG Spring Meeting*, Dresden, Germany.

J. Janssen, T. Hickel and J. Neugebauer. Automated convergence and error analyses for high-precision DFT calculations, *TMS Spring Meeting*, San Diego, CA, USA.

2016 **J. Janssen**, T. Hickel and J. Neugebauer. Automated convergence checks with the python based library pyiron, *DPG Spring Meeting*, Regensburg, Germany.

J. Janssen, T. Hickel and J. Neugebauer. Automated convergence checks with the python based workbench pyiron, *TMS Spring Meeting*, Nashville, TN, USA.